

REPORTING OF R&D

Following search is done to validate the requirements and components in natureScope project

PROCESSOR

We require a processor that can best for running a local LLM, an image classifier based on neural networks, a text to speech model and it also should have the capability to have ask me anytime mode as well as it should process the 4k camera images from its camera and render real time augmented reality objects on its screen.

1. Allwinner SoCs VS RK3566 & RK3576:

Allwinner: Here are some points that make allwinner **unfit** for our requirements :

To fulfill over requirement, we require dedicated NPUS, or AI accelerators but the allwinner lacks them both which causes efficient on-device LLM or image inference difficult.

Allwinner's lacks clear open-source licensing and doesn't integrate well with standard frameworks like VA-API or OpenMAX as it **uses CedarX**.

Offer weak CPU/GPU performance and low support for sophisticated tasks like AR or real-time inference

RK3566 & RK3576: Here are some points that RK3566& RK3576 **fall short** for our requirements

Rockchip RK3566:

NPU too weak: Offers only about 0.8–1 TOPS, barely adequate for lightweight inference, and insufficient for running LLMs or high-res image models.

Underpowered CPU/GPU: Quad-core A55 + Mali-G52—insufficient for heavy real-time workloads or AR rendering.

Unreliable UX: Community complaints about poor standby battery life, slow boot/resume, making “ask-me-anytime” challenging

https://www.reddit.com/r/SBCGaming/comments/1eshdib/rk3566_devices_really_dont_offer_a_good_user/

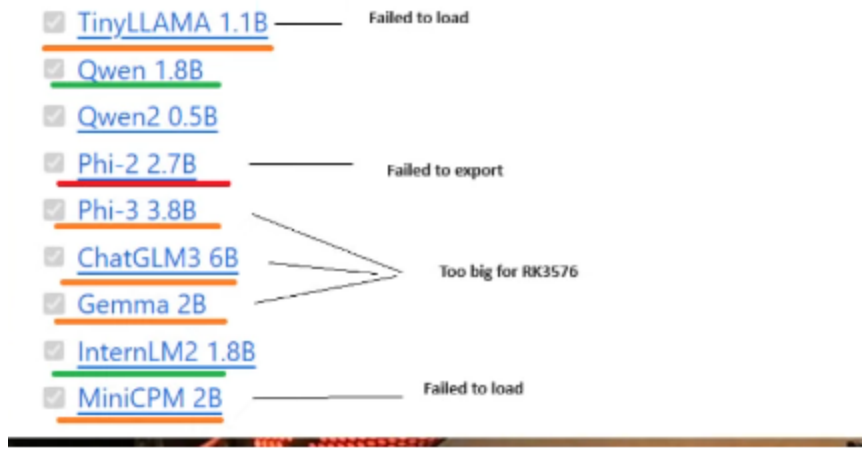
Rockchip RK3576:

Includes a more robust octa-core CPU (4×A72 + 4×A53) and 6 TOPS NPU, but this is still tight for simultaneous tasks like LLM inference, vision processing, TTS, and 4K AR rendering

<https://www.rockchips.net/product/rk3576/?utm>

Although capable of industrial AI tasks but RK3588 is far better for deployment of RK3576 in real-time AR applications. Due to some memory issue it doesnot support LLM models.

Support Models



<https://www.youtube.com/watch?v=RXDwdOtfog4&t=344s>

RK3576 can meet all of your requirements except AR rendering

Requirement	Capability	Notes
Local LLM Inference	Yes	RKLLM support; memory/NPU constraints exist
Neural Image Classification	Yes	NPU supports CNN-based models
Text-to-Speech / Voice	Feasible	Requires optimized audio pipeline
Always-On "Ask-Me-Anytime"	Possible	Needs low-power design
4K Camera Processing	Yes	Full ISP and multimedia pipeline

The price of rk3578 on LSCS is 23.04 USD

[SOURCE](#)

2. CAMERA:

The Sony IMX219 is an 8-megapixel image sensor used in camera modules for a variety of applications, including the Raspberry Pi Camera

Board V2, NVIDIA Jetson boards, and as a component in smartphones, action cameras, drones, and embedded systems. We can use Sony IMX219 but it is costly and out of our budget.

Sony IMX219 = 26 USD.

The Dex model uses 13 MP camera which is 13MP 4K Autofocus Camera Module CMOS IMX214 FOV 70 Degree Built in Digital Mic USB Mini Webcam For Windows Linux Mac Android

COST: 44 USD

Suggested camera:

1) Raspberry Pi 5 Camera IMX219 Sensor 8MP 77° Cameras Module for RPI 5 Jetson Nano 4GB 2GB + 15cm FFC

COST: 12 USD (better option)

[SOURCE](#)

2) Orange Pi 13MP Camera OV13855 Optional 80° / 120° FOV Fixed / Auto Focus Cameras Module MIPI CSI Webcam for RK3588 RK3588S

COST: 17 USD

[SOURCE](#)

3. MICROSCOPIC LENS:

To achieve magnification for close-up imaging, you need an M12 lens designed for short working distances and higher optical zoom. we have two options :

a) Arducam 16mm Telephoto M12 Lens:

focal length 16mm

Price : 8 USD

b) Arducam 25mm M12 Lens

focal length 25mm

Price : 18 USD

4. SPEAKER:

1PCS 2831/3128 Speaker 4 Ohm 3 Watt 4R 3W Full Range Cavity Sound Speaker HIFI Speaker Accessories 3525 4 ohm ultra-thin speaker

COST: 4.5 USD

[Source](#)

5. BATTERY:

3000mAh 3.7V 1S Rechargeable Lithium Polymer Li-ion Lipo Battery. The major consumers are RK3576 will consume 1.2 watts, and speaker will take 3 W.

COST : 3 USD

[Source](#)

6. GPS:

As already used in the prototype GPS01-TD costs = 5.78 USD, we can also use GPS ATGM336H which costs about 2.5 USD but takes more time to wake up and has problem with gaining longitudes and latitude making it less reliable.

COST: 5.78 USD

[SOURCE](#)

7.DISPLAY:

According to the prototype provided to us, we need a 2.8 inches TFT LCD with

Size: 2.8 inch

LCD Type: TFT

Outline Dimensions: 73*77.59*2.29mm

Active Area: 70.13*70.13mm

Number of Dots: 480(RGB)*480

LCD Controller: ST7701S

PIN NO.: 40P

Display Color: RGB

Backlight: White LED

Polarizer Mode: Transmissive

LCD Data Interface: 3-Wire SPI + RGB 18Bit interface

Storage Temp.: -30 ~ 80 degree

Operating Temp.: -20 ~ 70 degree

Viewing Direction: All

Voltage: 2.5V - 3.3V

COST: 16 USD

8.LED Ring :

led ring that will placed at the sides of TFT lcd will also have a diameter of 2.8 inches and its DC 5V 24-Bit RGB LED Ring, WS2812 5050 LED Ring for Arduino Raspberry Pi LED Display, with Integrated Drivers LED Ring 5V RGB Inner Diameter 2.8 Inches

9.

10.SUMMARY:

COMPONENT	PRICE (USD)	With Delivery (USD)	LINK
RK3576	23.4	37.5	SOURCE
Camera	13.87	free	SOURCE
Speaker	1.11	4	Source
GPS	5.78	23.03	SOURCE
Battery	2.48	3.37	Source
DISPLAY	16	25	SOURCE
LED Ring	7.32		SOURCE

Total Cost without Delivery: 70 USD

Total Cost with Delivery: 93 USD

IF we use TWO camera COST: 106 USD

